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Manifestos and manifestations: The role of a solar designer

Suzanna Törnroth

Design Research Group in Division of Humans & Technology, Department of Social Sciences, Technology and Arts, Luleå University of Technology, Luleå tekniska universitet, 97187, Luleå, Sweden

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ABSTRACT

What does it mean to be a solar designer? The article builds upon earlier work encompassing a 4-year study on the role of a solar designer, and offers further artistic and methodological explorations to re-define what it means to be a solar designer today. I autoethnographically explore what it means to be a solar designer through artistic material making at the world's first Solar Biennale, by performing 10 micro-phenomenological interviews with solar experts from different fields of expertise, as well as by conducting a workshop with 15 high school students as part of an Agenda 2030 exhibition at Norrbottens Museum in Luleå, Sweden. These designerly explorations, synthesised through an overarching framework of Relational Design, offer an updated perspective on what it means to be a solar designer – a necessary endeavour as part of an ongoing and evolving sustainable societal transformation. The updated visual taxonomy is offered as a guide to future solar designers and solar design researchers who are interested in a design-as-relationship-building approach, consisting of three key layers: *The Embodied Self, Space & Technology, and Embedded & Ecological Relations*. The article culminates in a discussion that expands and challenges these three key layers vis-à-vis discourses on sustainability, ethics, and relationality.

1. Introducing Solar Photovoltaic (PV) Design

> The urban dweller leaves their place in the morning. On the way to work, they pass by buildings with tilted structures and colourful glass facades – not paying attention to either. They reach their office and switch on their laptop. The ceiling and table lamp is on – electricity flows and supplies their daily work invisibly. They look outside and see a park with a children's playground. Caretakers sit under large shading structures to seek refuge from the bright morning sun. The urban dweller momentarily pauses their work to watch the children play, smiling, before being interrupted by a sudden phone call.

The narrative above exhibits the sun and solar energy in potentially different forms and meanings: the tilted structures, the glass facades, the lamps, the laptop, the shading structure, the phone call. In a fossil-free future, the common projection is that photovoltaics (PVs), in combination with other renewable energy, would generate the bulk of energy for daily consumption and use. The urban dweller's connection with renewable energy would expectedly extend beyond material sight and touch of the solar panel – solar energy would more widely impact what the urban dweller can *do*, in other words, their lifestyle choices and decisions.

For the scope of this article, solar design is effectively defined as the design of different types of PVs in place. By several scales of design – in both urban and dense settings as well as in open rural landscapes – PV technology manifests itself differently: as sprawling solar parks, building-integrated PVs (BIPVs), or standardised rooftop installations on buildings (Krstić-Furundžić et al., 2018). These technological forms are largely hidden from the pedestrian eye, which contributes to the phenomenon of electricity generation often being *out of sight, out of mind*. Global political frameworks, such as the United Nations' Agenda 2030, offer guidance for businesses, organisations, and governance authorities worldwide to adhere to strategies towards a fossil-free future. Though, at what cost? Research has indicated prevailing conflicts and tensions surrounding solar design (some different, and some in similarity with wind power design): myopic and ineffective use of lands from the perspective of long-term land-use planning (Scognamiglio, 2016), major metal extraction and mining in high-risk contexts (Lèbre et al., 2020), decolonialism and indigenous solidarity against solar (Kinder, 2021), Not-In-My-Backyard Agenda (Devine-Wright, 2009), efforts and contentions within energy democracy and accessible community *prosumer* design (consumers who produce their own electricity) (Stephens, 2019; Van Der Schoor & Scholtens, 2015), the challenge of scale and landscape-technology compatibility in aestheticized spatialities (Phadke, 2011), and lastly, a resounding tension of, "[w]hose aesthetic preferences will we follow?" (Caffrey, 2010, p. 742).

This urgently calls for a designerly solution where a co-creative

E-mail address: Suzanna.tornroth@ltu.se.

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approach that delicately weaves together the (many) fabrics of society could set the foundation for sustainable PV design. This article delves into the research statement of *What it means to be a solar designer, redefined?* through a series of artistic and methodological explorations. An artistic exploration positioned myself, as the author and a solar designer, to autoethnographically explore human-sun relationships through material making at the world's first Solar Biennale. Following that, I conducted 10 micro-phenomenological interviews with solar experts from different knowledge fields: municipal urban planning, politics, design consultancy, architecture, and environmental engineering, in order to holistically capture the experiences of the solar transition in society. Finally, I held a workshop for 15 high school students as part of an Agenda 2030 exhibition at Norrbottens Museum in Luleå, Sweden. The workshop focused on engaging local youths to investigate their values and ambitions towards sustainable living in Sweden. These explorations are then analysed to improve and re-define a taxonomy created on the solar designer as part of my dissertation (see Fig. 1) (Törnroth, 2023). The re-visitation of this earlier work allows for the role of the solar designer to evolve and develop as part of the ongoing sustainable societal transformation – a transformation that is always shifting and moving in focus.

2. Background: Defining the Solar Designer

The term "solar designer" has been a recently coined designation upon designers that have begun to identify themselves with the aesthetics and craft of solar PV panels in different forms. Creative experts within the field – Pauline van Dongen and Marjan van Aubel specializing in artistic solar PV design among other things – are paving the way to putting solar design into action. Culminating 4 years of multidisciplinary doctoral studies in architecture and design, is Fig. 1, a taxonomy depicting the role of the solar designer (Törnroth, 2023). The taxonomy gives a brief overview of what it means to be a solar designer designing for PV technologies in the modern world.

The role of the solar designer entails four categories: *The Design (Research) Process, Human Experiences, Space & Technology, and Embedded & Ecological Relations*. These categories were built upon the phenomenology of solar energy technologies (particularly, the PV panel), as being situated, relational, and centered on the human experience. The categories do not have boundaries between themselves because their separate sub-traits might overlap and shift depending on context. A description of the different categories is as follows.

The Design (Research) Process explores solar design as a process and a product, and the potential impacts on the wider world(s). Designerly paradigms, teachings, and teams within solar design have the capacity to challenge the status quo in a transformative, action-oriented, ethical,

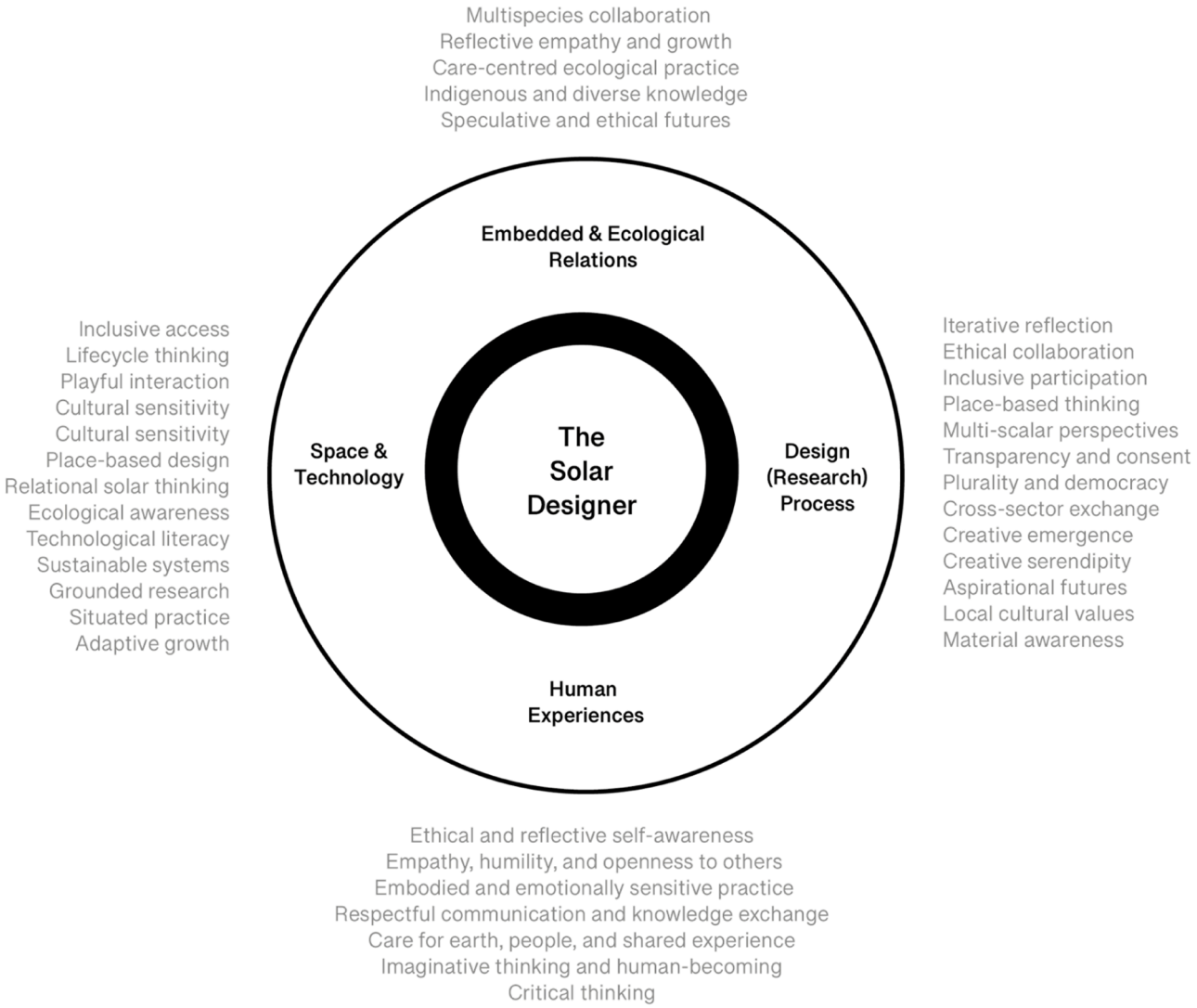


Fig. 1. Taxonomy of the solar designer.

situated, and collaborative way, to produce meaningful outcomes for as many as possible. The category *Human Experiences* reminds us that the designer is always first, and foremost, a human being, embodied with subjective thoughts, feelings, senses and more. Honing an appreciation towards a certain noticing (Tsing, 2015), alongside traits such as adaptability and curious learning, may sensitise the solar designer to different human and non-human narratives that might spark wonder, inspiration, and passion for change. *Space & Technology* signals a need for the in-depth know-how towards the techno-economic structures and systems at play in the global context, in order to cleverly navigate contextual sustainability and energy efforts. This includes understanding the basics of the technology in question, certain political and economic relations and forces at play (e.g. the consumer market, power, and monopolies), and different legal, environmental and spatial parameters (e.g. land-use zoning and ownership, forgotten and marginalised histories, private versus public spaces, and more). Papanek (2019) aptly explained that, "[t]he most basic study of system design shows that a system made up of component parts will change eventually as each part is changed" (p. 275). Finally, the category *Embedded & Ecological Relations* emphasises relationality in understanding how solar technologies are positioned within human and non-human lives. Awareness of human's reciprocity and *intra-actions* (Barad, 2008) between these worlds broadens current solar energy discourse, which is paramount in an occupation as anticipatory as the solar designer, and a movement so rapidly evolving as solar design.

3. Method

Burzyński and Burzyński (2014) characterised being uncertain in post-industrial modernity as a general distrust towards technology. They connected this to Ulrich Beck's conception of the term "risk society", which refers to the negative consequences induced by processes of social change:

> In this context, the notion unveils its typically cultural function as a kind of discourse, a certain ideologically-tinged way of perceiving reality in which technology and modernisation are dismissed as potentially destructive to individuals and communities. Consequently, the idea is very often seen as a sociological shorthand for criticising contemporary technological civilisation; it constitutes a new vernacular of public concern stating that the West and the western model of living have probably led us astray (p. 27).

To design for uncertainty means to be creative and resourceful under impending restrictions – that is, to design for resilience, adaptability, and reuse. Solar PVs as they are designed today, are anything but. By examining their production chain, one can understand that these panels follow a linear and extractive production culture. In 2021, only about 10% of panels in the US were recycled (Crownhart, 2021), even while an estimated value of the metals in all PV installations can approximately amount to 22 billion dollars (Domínguez & Geyer, 2019). Currently, it is much more expensive to recycle them as opposed to simply discarding them. The average life- cycle of a solar panel is approximately 25 to 30 years, and by 2030, the predicted cumulative mass of end-of-life PV is projected to total 8 million tonnes globally (Heath et al., 2020). By 2050, this mass is projected to increase to 78 million tonnes, contributing greatly to the global production of electronic waste (e-waste) (Chowdhury et al., 2020). Thus, waste from solar panels can be considered the fastest growing e-waste component globally (Majewski et al., 2021). In addition to the mining of toxic metals for PVs continuation and the profuse manufacturing at fossil-based industries, PV panels are far from being green in themselves. However, they are symbolic of a green transition for the reason that as a functioning product, they can be purchased, handled, and installed with ease, due to their generally low costs, modularity, movability, and open possibilities for off-grid and on-grid systems, with little to no greenhouse gas emissions in the process.

The method for this article is thus one that builds on the taxonomy of the solar designer, exploring further how uncertainty in PV futures might influence how *(should)* the role of the solar designer be changing. The method encompasses a personal introspection event (to delve into the *humanness* of being a solar designer), 10 micro-phenomenological interviews (to expand on the categories *Space & Technology* and *Embedded & Ecological Relations*), and a workshop with 15 high school students (focusing on this article's research question qualifier – sustainability futures and uncertainty). This curation of methods forms the basis of a sort of relational design approach (Udoewa & Gress, 2023): "[w]ork in relationality in design… contributes to an entire field of critical self-reflexivity for designers aimed at advancing design justice" (p. 105). Relational design (RD) argues that an awareness and focus on relationships affects all the other aspects of a system. Instead of seeking system health by trying to change power, resource flows, or purpose, it might be more beneficial to focus on system relationships:

> As relationships deepen and the number of deep, healthy relationships grow, resources will flow. Power then shifts due to changing resource accumulation and flows and shifting relationships between those with more power and those with less power. Ultimately, new deep relationships along with resources and power can change the purpose of the system as well (p. 122).

Thus, the power of RD is in slowly shifting systems by building relationships, creating resource flows, changing power, and consequently altering the purpose of small (sub)systems. RD transforms traditional (colonialist and extractivist) information gathering, ideation, and processes of making in design by utilising relationship-building methods and the design-as-relationship-building model. For better understanding, a story is shared within the article by Udoewa and Gress (2023) in order to depict this idea situationally:

> An example of a successful RD project is a 12-member community that used a mix of intentional communal living (a few members) and weekly community meals, sharing, and discussion (all members). Through our relationship building, we learned about passions for working with children and a burden for homelessness. We organically grew a shared desire for relational service in those areas. We designed community projects that we did alongside young boys at a home for orphans. We also designed relationship building time and space with the young boys. Through those relationships, knowledge of the boys' situation, countries, and families flowed to us, and our resources (time, bicycles, funds, etc.) flowed to them. Over time, the relationship transformed from one of a group of benefactors and orphans, to contributing members of our city. One boy, now a man, was able to attend university and find a professional job. Another started a business. Another became a physical trainer. There are many similar stories (p. 122).

This approach is applied in this article through the appreciation and seeking of knowledges beyond mine, in a participatory, open and meaningful way, that evolves and directs itself based on mutual sharing and trust. The culminating RD experience is summarised in the Discussion.

4. From the Introspective Self: Solar Manifestations

The world's first Solar Biennale occurred in 2022, in the Netherlands, between the months of September to October. A program packed with seminars, lectures, tours, pop-up workshops, performances, and art exhibitions engaged the question: *How can we design a post-fossil future using the power of the sun?*

I participated in a four-day Solar Lab on the topic of The Personal, where the participants were challenged to conceive a probe that facilitated embodied interactions with solar technology, to give new emotional and aesthetic values to solar that would make it more approachable, tactile and inviting. By way of dissipating the researcher-

user dualism (Xue & Desmet, 2019), I engaged in an autoethnography endeavour – a first-person, researcher introspection that would enable myself to immerse in and express subjective experiences from this privileged access to a peculiar and specific event, which is not directly accessible from a third-person perspective. This autoethnographic positioning does not merely document experience but operates as a method of inquiry in itself, where embodied perception becomes analytic material. In doing so, it allows tacit, affective, and pre-reflective dimensions of engaging with solar energy to surface – dimensions that would remain inaccessible through detached observation alone. According to Byrne (2012), "[*Knowing*] has the interesting property that, necessarily, if one follows it, one's resulting belief about one's state of knowledge is true," (p. 190). Thus, my participation in the Solar Biennale was categorically one of, "readiness, vividness, richness, and depth, " (p. 41) where I consciously observed the embodied sensory feelings, perceptions of meaning, and emotional reactions from the process – bringing into focus the humanness and the embeddedness of my being within a larger ongoing meta-process.

The Solar Lab entailed integrative workshops from design researchers and practitioners, pauses for meditative acts on conscious bodily energy flows, and lectures from solar technology experts. The unique perspective on the Lab was that it moved away from engaging with static forms of solar PV technologies and instead immersed into speculative solar conceptualisation and manifestation; the perception of different forms of human-sun relationships that challenge conventional solar PV design. The Lab refocused on the role of the sun as the provider of an energy system that supports and sustains life in meaningful, immaterial or material, individual and subjective, ways.

According to the activities of the workshop, I created a probe with the intention of celebrating the sun's energy as it so presented itself in the current seasonal environment in late October 2022 in Eindhoven, the Netherlands: a cool and crisp breeze, darker days, crunchy fallen Fall leaves, and a yellowing and reddening of nature. The probe (*refer to Figure 2 in supplementary materials*) was thus a table runner created of fallen leaves and wasted cloth material in the colours of the sunset, sewed together and on a backing of ironed cellulose fabric. The flowers were made with wasted fabric sourced from older projects within the workshop, hot glued on to the table runner. The leaves, positing as a material capturing and celebrating the sun's meaning and significance in that point in time, were produced by a biological process coaxed into being by the shortened hours of sunlight during the day and the lower temperatures in the Fall. This biological process stimulates the release of a hormone called auxin in trees, weakening the attachment of leaves to branches, thereby resulting in falling leaves off branches, to prepare for minimising water loss by the tree over the cold winter months. The firmly present, yet temporary and organic nature of the leaves that can be easily obtained from sidewalks, offers one of many ways human-sun relationships might manifest.

The tactility of the runner – the crinkling sound sharply exuding from the piece as one places and removes items from it – and the soft and lasting earthy smell of fresh autumn air that wafts from it, was intended to serve as a reminder of our – humans and humanity's – situatedness within our surrounding environment and its seasons. The table runner celebrates how the sun shapes seasonality and therein, our experiences of the world around us. As aptly put by Miller et al. (2015): "Energy shapes—and is shaped by—people's economies, workplaces, identities, environments, technologies, landscapes, politics, and mental maps of the world," (p. 38). Such human introspection and reflectivity over our own personal relationship with the sun (and solar) extends beyond the role of craft as a designer – allowing us to focus on the humanness of daily experiences and how that might shape perceptions, meaning-making, and sense-making of sustainability in our situated environments. Deeper appreciation for natural and biological processes heightens the centrality of human entwinement with nature – dissolving notions of humans being apart, or separate to, experiences of weather, climate, and the environment (Neimanis & Walker, 2014). Succinctly explained by Alaimo (2010) in *Bodily Natures*:

> Indeed, thinking across bodies may catalyse the recognition that the environment, which is too often imagined as inert, empty space or as a resource for human use, is, in fact, a world of fleshy be-ings with their own needs, claims, and actions (p. 2).

5. To Phenomenological Knowledge: Solar Manifestos

Beyond an embodied and situated individual experience, I posit that the role of a solar designer necessitates a multi-scalar, temporal, and plural understanding of the solar discourse and transition. Micro-phenomenology is a qualitative interview method that focuses on gathering a second-hand account of a first-person perspective on certain lived experiences (Petitmengin, 2006; Prpa et al., 2020). The utilisation of micro-phenomenological interviews was thus to holistically understand and centralise the interviewees' experiential content and processes of the solar transition. By way of 10 in-depth interviews with solar experts from different knowledge fields: municipal urban planning, politics, design consultancy, architecture, and environmental engineering, I intended to capture the experiences of the solar transition through the lens of different experts in different contexts, roles, and knowledges within the transition, reflecting the aforementioned categories of *Space & Technology* and *Embedded & Ecological Relations* in Fig. 1. The motivation was to explore plurality in perspectives and nuances in the solar discourse, interrogating the harmonious public imagery often depicted in global media towards the future clean energy transition (Hickel, 2019). Values and priorities differ across people – and by values and priorities, I refer to the perceived significant dimensions and qualities of the solar transition (by the individual or their organisation), that give meaning and ultimately drive their work. The interviewees were initially selected via purposive sampling (Flick, 2018), while their roles and expertise were subsequently characterised from the bottom-up – in other words, they were probed to clarify their own perceived scope of expertise. The interviewees are anonymously delineated in Table 1.

An interview guide of approximately 8–10 questions was established to guide a sensitive unfolding of all the complexities of experiential content:

- Scope of individual and organisation's work and profile.
- Motivations behind their work with energy, climate, and the environment.
- Challenges faced in their line of work.
- Future opportunities they perceive for renewable energy in society.
- *Their future energy utopia*: What would their ideal energy future be like?

The interview guide was adapted for each interview, and thus, the exact number and type of questions varied. A thematic content analysis framework was used to assess the values and priorities surfacing in the interview transcripts (Braun & Clarke, 2006). Codes of interest were first identified and collated via multiple readings of the interview transcripts. The codes can be characterised as words or short phrases where "something of special interest" (Simonsen & Friberg, 2014, p.111) was observed. The codes were then categorized into themes. The themes (values) that evolved from the transcripts were: *Manageable Investment Costs, Bottom-up Flourishing, Cross-sectoral Collaboration, Context Embeddedness,* and *Aspirations*. Table 2 summarises characterising quotations from the different interviews and how they connect to the different themes.

5.1. Manageable Investment Costs

A central topic in all the interviews was cost – cost surrounding access to solar energy, its material infrastructure and its services. The topic

Table 1
Details of interviews.

Interviewee Code	Identity and Expressed Expertise	Information Collected (with Consent)
Participant 1	Citizen health and well-being strategist in a Swedish municipality.	Audio recording
Participant 2 and 3 (interviewed in a pair)	Global start-up duo in artistic renewable energy design (an artist and an architect).	Audio and visual Zoom meeting
Participant 4	Architect working closely with citizens and grounded implementations at a non-profit in Singapore.	Audio and visual Zoom meeting
Participant 5	Urban researcher researching socio-cultural development with marginalised communities in London, UK.	Audio and visual Zoom meeting
Participant 6	Environmental engineer studying smart technologies and their impacts and ethical implications on social life in Finland.	Audio and visual Zoom meeting
Participant 7	Product and service designer in a company designing urban street furniture integrated with renewable energy technologies and selling globally.	Audio via a telephone call
Participant 8	Politician-activist in environmental efforts in Alaska, USA.	Audio and visual Zoom meeting
Participant 9	Founder of a start-up in California, USA, that facilitates community ownership over decentralised renewable energy systems through the creation of cooperatives.	Audio and visual Zoom meeting
Participant 10	City councillor in Sasketoon, Canada.	Audio and visual Zoom meeting

of cost manifested in different forms: discussions regarding different political initiatives and subsidies, the solar panel consumer market, cost of access to cheap electricity and for whom (see background reading on just electrification (e.g. Tarekegne, 2020)), as well as the cost of privacy. The cost of privacy is reflected in the discussion by the amount of information collected from consumers during the installation of smart energy home systems.

These challenges presented themselves as burdens of the solar energy transition, in which all the interviewees generally agreed to this hypothetical statement: the cheaper the better. However, upon examination of the quotations in Table 2, this view becomes problematic when shared by all; the politician who is scared to raise taxes for fear of losing public favour in the next election at the expense of infrastructural maintenance, care and well-being in the city, a homeowner who would like a cheap utility bill at the expense of indigenous others, or the product designer and developer who seek out profitable materials and markets at the expense of the environment. To some extent, these arguments call for the redistribution and renegotiation of costs across actors in the solar energy system. Participant 3 discussed the idea of "transcending silos", where costs could be managed lucratively when cross-sectoral collaborations occur in solar energy development projects (Participant 3, 2020). Such arguments afforded the consequent deliberation of costs and their significance in context. In other words, costs become re-evaluated from a different perspective, challenging the currency of doing solar energy. One example could be the idea of ecological accounting, where socio-environmental factors are integrated into reporting, performance appraisals, and decision-making processes (Birkin, 2003).

5.2. Bottom-up Flourishing

The interviews indicated the need, and importance for, local inhabitants – in some interviews, specific mention towards indigenous peoples – to be involved in the design and implementation of solar

Table 2
Themes and characterising quotations.

Themes	Quotations
Manageable Investment Costs	"Certainly the neo-liberal brand, we're never going to make real progress unless we mix structural change in the way that we incentivise investment." "Like everybody agrees that we want to do renewable energy, [but we also] want to save money on our electricity bill." "Politicians were so scared to raise taxes so they didn't do a proper renewal on the basic infrastructure like roads and sidewalks, and in the last 10 years there's been a process of catching up."
Bottom-up Flourishing	"It has become clear to me that we need to be a lot more organic in the way that we engage with the community and let them come up with ideas as they come up." "One thing we do is, is less knowledge from us, but meeting people from different disciplines, networking them and connecting them to hear from each other and learning from different methods of each other." "The first thing is that we really don't work in communities unless we're invited in... We don't want to impose ourselves. It needs to be organic and really nurtured from the bottom-up in the community." "So much [indigenous] culture is embedded in a better environmental process and yet they are sort of left out – like they don't have a lot of control or power on almost any level of government other than their own."
Cross-sectoral Collaboration	"It feels like it's really more about relationship building, and I'm trying to listen and design something that fits that. It feels like people are more interested in seeing us being transparent about our design process and taking in some of their key tenets." "You can't only have "open invitation" sort of dialogues because most people that come then are white, 60-year-old men, so the municipality sometimes uses an interest-mapping process for developments."(This text has been translated from Swedish) - The biggest challenge is in the "regulatory obstacles" that hinder cross-sectoral innovation. - "Match-making different skills and needs" is needed. *(Regarding navigating experts' egos in collaboration)* "Sometimes we need to advise them to take a step back and listen, because what the locals are saying may help a lot as well."
Context Embeddedness	"They really didn't want to convert to wild land to develop. They were not excited about that. If we were going to do that, they were very insistent that we do dual land use, like put agriculture underneath it, so we won't be relegating it to sort of an industrial site." The company distributes their products to over 26 different countries worldwide, each with their own specific climatic context – from dry deserts to the arctic cold. This puts a pressure for their solar PV products to operate in a wide range of temperatures. "[W]e have three generations that are present and economically active, we have people in that 20 to 25 years old, their parents and their grandparents. Their approaches to technology, and even their philosophical approaches to what using energy means, differ greatly." "I have found that some of the value of the project is that it's intimate and we build community and relationships through it. …They started a weekly community potluck dinner as a result of enjoying our monthly meetings. We would just come to their weekly potluck dinner, also sometimes talking about solar. But mostly not."
Aspirations	"So we don't need to see infrastructures as objectives. Just as processes in enabling – how we enable people, and in that, is infrastructure." *(Regarding smart systems)* "I philosophically speak about it – kind of like a smart butler. A butler that knows everything about you and he's able to anticipate your needs. … The butler is a faithful butler to you. At the door, it just prevents any intruder to get to know your details. And the butler says this is how much we need, and this is when we need it. So where do we find this trade-off? What does the butler need to tell the other side that and for what reason?"

energy technologies. The word 'involved' here, encompasses several levels of involvement – from participating in citizen consultation processes, to heavier responsibilities of leading and guiding development processes (see Arnstein's (1969) ladder of citizen participation). Several words that struck as important from the interviews in characterising the idea of bottom-up flourishing are, "nurture", "organic", "learning" and "power". These words imply key tensions that exist – which influence the relationships and processes that shape solar energy transitions – between those who are assumed to have expertise in the matter (e.g. scientific research community, politicians and practicing engineers) and those who are assumed to not have expertise in the matter (e.g. citizens assumed as a "homogenous whole", children, and students).

The question thus arises: what is understood (assumed and projected) as expertise? What type of expertise is being referred to, and how is it formed? The interviewees leaned towards the idea that a certain lived experience of the context and space provides local inhabitants a level of invaluable expertise upon which they refer to in advising the (most optimal) ways of designing and implementing solar energy in a specific spatial environment. This expertise can be understood as a deep sense of attachment and belonging to the socio-cultural aspects of a space (Green, 1999), as well as the experiential knowledge of the long-term and inner workings of the environmental, climatic, and spatial aspects of the space (Lindberg et al., 2020). The interviews signalled the prioritisation of this type of expertise, as fundamental knowledge in jumpstarting a design and implementation process within solar energy.

5.3. Cross-Sectoral Collaboration

Not unlike the previous theme of bottom-up flourishing, key words and phrases that have been identified from the interviews to characterise the importance of cross-sectoral collaboration link to the nature of interpersonal relationships: "listen", "match-making", "relationship building", "interest mapping" and "transparent". The previous section indicated words that implied growth and development from the bottom-up, while the words in this section imply the transversal and ethical dimension of human-to-(more-than)human connection, regardless of assumed expertise, role or responsibility. The aforementioned words point to a call for sensitivity, humility, and empathy in the collaborative process, beyond the necessary and pragmatic complementing of skills and knowledge between different experts. The interviews signalled that the constellation of actors in a solar energy transition matters – but what seems to matter more is *how* the constellation communicates, learns and develops together and from one another. Previous research in the field of ethical practice within research communities supported this notion by explaining that: "Only open dialogue can develop the true ethical strength of the research community and shape its identity in society" (Mustajoki & Mustajoki, 2017, p. 61). Beyond the case of a scientific research community, the interviews showed that this statement resonated across all sectors of society.

5.4. Context Embeddedness

This theme emerged from the characteristic use of words such as "specific", "intimate", and "present". These words pointed to the immediacy of local particularities (context) in a coalescence of time, place, things and people. The interviews showed that tensions arose when these local particularities do not align with top-down initiatives in solar energy. As an example, Participant 9 explained that in her experience, tight communities often have distrust towards external developers. In Hawaii, where she conducted immersive groundwork for half a year, she explained that indigenous peoples have had a long history of local land exploitation for meaningless and profit-oriented developments in the name of tourism. This has resulted in hostile behaviour, where these communities are now forced to protect their lands by becoming upfront with developers from the beginning: "… asking where they are from and how much money they are going to make off this project". Solar energy infrastructures in these cases can often become political and symbolic – from its process of development to its aesthetic on a landscape. Another example from Participant 9 is the case of Fresno, California, where she explained that a tight community shares a strong sense of care, obligation and ownership over the peripheral wild lands: "… they were very insistent that we do dual land use, like put agriculture underneath it, so we won't be relegating it to sort of an industrial site" (Participant 9, 2020). Thus, context embeddedness as a value focuses on the importance of fair and transparent decision-making at all levels of solar energy governance, where inclusion and diversity of different social groups through iterative communication is needed to build trust.

5.5. Aspirations

Across the interviews, the question pertaining to the interviewees' depiction of utopian future energy imaginings sparked a consistent elusive conceptualisation of "*getting there*" – "*there*" often being considered as a utopian end-goal that society should or could achieve; not an alternate world or non-place imagining. But, what is this elusive "*there*" – the future, a potential value? The question relates to what future innovation can be founded upon in the context of today – concerning the actions and decisions multiple actors can make today – in order to get "*there*". The interviewees consistently discussed the issues and challenges of today that make it difficult to get "*there*"; the ideal renewable energy future that is hoped for. This was understood through several identifiable words and quotations, with examples being, "processes in enabling" and "anticipate".

One interview pointed to a future need for the increased "smartness" of a system, and the need for added flexibility in the process of the design and implementation of renewable energy infrastructures. Another interviewee argued that bureaucratic issues associated with governance and law act as hindrances to the growing flexibility of renewables' systems. A third interviewee explained the challenge of how a narrow or limited mindset might impact what infrastructures are – *and what they could be*. This interviewee explained that value should be placed on the time-space perspective of infrastructures: "What happens to the places before and after it is built? Do infrastructures take on different meanings and values across the span of time?" (Participant 5, 2020). She offered that infrastructures had a temporal meaning – that the making of infrastructures should be considerate to its multiple forms of existence (and legacy) across time and space.

The interviews offered that a certain level of reflexivity is needed in designing for solar energy – and the renewable energy transition as a whole – because humanity cannot yet fully comprehend the multitude of long-term impacts these infrastructures have on both human and non-human living environments until they are built, maintained or degrade, and then eventually demolished. Today, humanity might be working towards renewable energy futures. Tomorrow, another newfound scientific revelation might take its place. Infrastructural meaning and function are necessary – but their forms, *fleeting*.

6. Youths – The Future? Reconciling with Uncertainty in Ways Forward

As explained in the Method section, the purpose of this workshop was to expand further on the role of the solar designer through an additional qualifier – being a solar designer *in uncertain times*. In March 2021, I held a sustainability workshop at Norrbotten Museum in Luleå, Sweden, as part of their wider exhibition "We are the World" (*in Swedish: Vi är planeten*). The exhibition's motivation was to promote the learning of the Sustainable Development Goals (SDGs) as part of UN Agenda 2030, and how these goals manifest themselves within the local context of Sweden. The motivation behind the workshop was to engage local youths to investigate their values and ambitions – as well as challenge their knowledge – on sustainability matters in Sweden. A class of

approximately 15 high school students studying tourism and hospitality attended, along with their teacher. These students ranged in ages from 16 to 19 years old. The workshop was conducted digitally on a collaborative Miroboard because Covid-19 restrictions had prevented a physical gathering.

Since it was difficult to translate the aim of rich physical interaction with solar from a planned interactive exhibition into the medium of a stripped, two-dimensional experience of the computer screen, I shifted the focus of the digital Miroboard workshop to one that pertained to personal knowledge, experiences, and relationships with sustainability matters. The digital Miroboard workshop consisted of two activities that invited the engagement of youth on different aspects of sustainability matters.

Activity 1 of the workshop was a "Sailing Boat" exercise, where the participants were encouraged to explore their thoughts and values on the question: What drives society towards a future only using renewable energy? (*translated from Swedish – refer to Figure 3 in supplementary materials*). Dot voting was encouraged – the act of placing red dots on ideas that are preferred – and some of the popular answers included: "positive attitudes", "innovative people", "creative ways of thinking how renewable energy could be", "to push on", "information", and "fluent skills". Other answers included: "research", "education", "technology", and "cities that we are passionate about". The participants were then encouraged to tackle the opposite question: *What hinders society from achieving a future with only renewable energy?* Dot voting shows that popular opinions included: "lack of knowledge", "bureaucracy", "being too comfortable in old ways", "high costs", and "slow decision-making". Other answers included: "large-scale application", "poor technology, populism, overpopulation, and low motivation".

Activity 1 indicated a certain level of skepticism towards a renewable energy future, judging from the higher level of engagement on the second question than the first. The popular belief is that governing and education structures in place do not currently invoke confidence towards the attainment of a future with renewables, due to challenges of "bureaucracy" and "slow decision-making", as well as the "lack of knowledge". Noticeably, there appears to be a clear division across the participants, with some perceiving that educational structures currently play an effective role in contributing to "fluent skills", "information", and "innovative people" in Swedish society. While there was no direct engagement between the different perspectives, the range of answers signals diversity in thought. Furthermore, the diversity in thought signals a reliance on – and a glimpse of hope towards – top-down structures to facilitate the renewable energy transition, with most opinions targeting the roles and responsibilities of larger authoritative structures and their perceived successes and failures. It appeared that within this context, importance was placed on seeing quick and flexible decision-making, creative and plural solution-finding, and motivated behaviors towards sustainable change.

In Activity 2 (*refer to Figure 4 in supplementary materials*), the participants were then encouraged to rank the values and ambitions they shared in Activity 1, according to the two axes of importance (low importance, medium importance to high importance) and time (important to act on now, or only important to be acted on in the future). The motivation behind this exercise was for the participants to consider and assess both the importance of their suggested perspectives, and when these needed to be acted on, in a bid to achieve a renewable energy future. In line with Schoor and Scholtens (2015), this was intended to challenge them to, "turn these ambitions to reality" (p. 667). The results showed most answers being considered as high importance, but of average urgency to be acted upon. These answers include: "curiosity and research", "fluent skills", "high costs", "innovative people", "being too comfortable in old ways", "good education", "technology and research", and the all-encompassing answer, "everything is important". Another answer that was deemed more urgent to act on, while of medium importance, is "positive attitudes".

Through the activities, the participants clarified what they deemed was important to achieve in Sweden's journey towards a renewable energy transition – with these values and ambitions naturally isolated to their contextual knowledge and experiences. Overall, the results showed that they might not feel currently supported by overarching governing and educational structures that can facilitate such change, and that these organisations' responsibilities are not presently fulfilled. To reiterate, these results, although situated and context-dependent, follow a phenomenological approach and thus still offer a meaningful and fruitful platform for further discussion and deliberation over the role of a solar designer.

Hickman (2020) explained about the notion of eco-anxiety and how it plagues masses of youth in recent times, where, "young people [are] not knowing who they can safely talk with about their anxieties about the future" (p. 413). These findings hint at the distress that these youths might be experiencing – frustration at how governing bodies currently handle the sustainability transition, whilst still hopeful for a future where change and prosperity might be imminent. I offer a segue into a discussion on how – and what – a solar designer is, deliberating and reconciling plurality on this very matter: Why is humanity at conflict as to how the sustainable energy transition is managed? Who dictates what is sustainable, and how does that impact different contexts? In the words of Hickman (2020) when she asked youths what they wanted from adults, "they consistently said that they wanted adults to be brave and to 'tell the truth'" (p. 420). Perhaps this is the true calling for a solar designer: to be transparent, to adapt with empathy, and to critically assess a renewables transition in-context.

7. Results & Discussion

So, then what does it mean to be a solar designer? The recent solidification of this realm of knowledge within a single identification calls for the questioning beyond *what it means* to be a solar designer, but rather, *how*. How to be a solar designer first and foremost, includes an acceptance of being human, for the reason that it anticipates an occupation that might emerge as a necessary part of designing with and for human (flourishing in) living environments. Resounding in the words of Papanek (2019): "Yet designers *are* involved, like it or not, as human beings" (p. 278). With the added qualifier of *how* to be a solar designer in uncertain times, the updated taxonomy aims to emphasise traits that represent resilience, adaptability, and openness towards and curiosity for change. This synthesises and responds to knowledge derived from the personal introspection event, the micro-phenomenological interviews, and the workshop. These three explorations were founded upon the older version of the taxonomy, to birth a newer and more relevant depiction (Fig. 5).

Fig. 5 shows that the basic traits of being a solar designer remain the same, however, the framework of thinking surrounding the role of the solar designer has changed. Previously, heavy significance was placed on the solar designer themself – shielded-from and othered-than the other four categories. New knowledge shows that the solar designer is – while central – is only but a small part of the larger, more encompassing universe of *Embedded & Ecological Relations*. The novelty in the proposed update of the taxonomy is thus the uncovering of the transitioning significance from the solar designer towards the universe of *Embedded & Ecological Relations* in which the solar designer operates in This reflects the reasoning behind the renaming of the solar designer as well, to *The Embodied Self*. The solar designer, while important, is not necessarily identified and prioritised solely by their occupation, but rather, by their humanness and how they perceive and engage with the world(s) around them. Their occupation is a complement towards their core self and philosophies: sensing, understanding, synthesising information around them. The power of its language, *The Embodied Self*, is necessarily noticeable: as an earthling, the solar designer exists alongside other earthlings.

The porous boundary between the solar designer, now called *The Embodied Self*, and the rest of the framework indicates a fluidity and

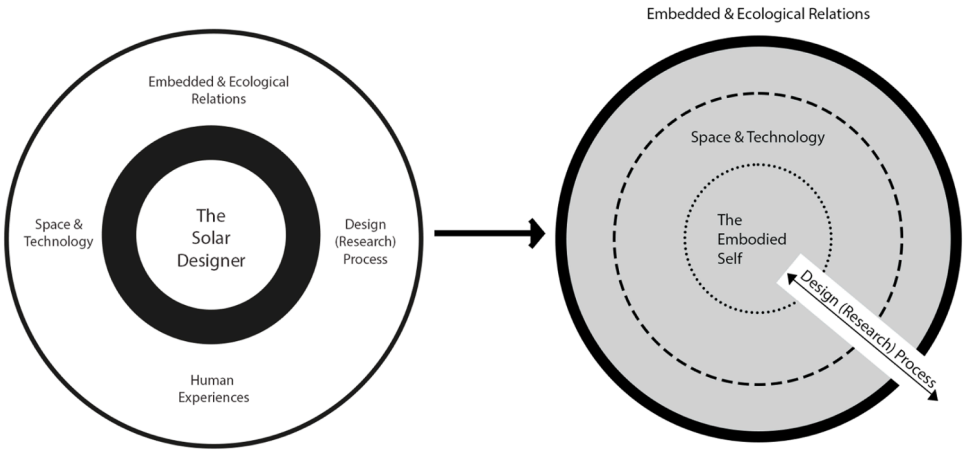


Fig. 5. A proposed update of the original solar designer taxonomy in Fig. 1.

relationality between and within the ecological system. *Space & Technology*, characterised by material solar energy technologies and immaterial influencers such as legal, environmental, and spatial regulations, lies in the middle ground that buffers and surrounds the space that is taken by human and non-human bodies. *Space & Technology* is a regulatory layer that disciplines and directs how *The Embodied Self* positions itself and operates in the system of *Embedded & Ecological Relations*. Often techno-economic in nature, these are forces that the solar designer might not be able to escape – but might be able to challenge in certain respects – as indicated by the thicker and less porous (but nevertheless still porous) lines. Finally, the design process should not have been an isolated category on its own. The design process extends across all layers and boundaries, bridging different categories through messy, iterative, and relational thinking and making. In the lens of relational design (RD), this, "requires an awareness of power dynamics that can undermine healthy relationships and caution when engaging as a professional designer" (Udoewa & Gress, 2023, p. 122). The design process, based on design-as-relationship-building (reflecting the RD mindset), focuses on the interrelations and *intra-actions* (Barad, 2008) between *The Embodied Self, Space & Technology*, and *Embedded & Ecological Relations*. This is learned through the introspective piece, the interviews, and the workshop, as explained further in the following paragraphs.

Being a solar designer entails a wider questioning beyond solar technology in itself, outwards into different worlds and knowledges that provide plural inspiration for how this craft can and should take shape. Different values present in these different knowledges may contribute to tensions and conflicts in the solar design discourse – all of which the solar designer might need to consider in the performance of their craft. However, I believe the situatedness and subjectivity of value perception and interpretation is significant – that *the clean energy transition can look different in different parts of the world*, contrary to how global media and politics often portray the renewables trend through hegemonic imagery (Hickel, 2019). The interviews show that certain values may be considered consistently significant across different knowledge groups: *Manageable Investment Costs, Bottom-up Flourishing, Cross-sectoral Collaboration, Context Embeddedness*, and *Aspirations* – all of which feed into informing how it is to be a solar designer in uncertain times. A broader interpretation of the interviews and the introspective piece may lead to more comprehensive points of departure for further discussion and debate:

- Manageable Investment Costs: Certain techno-economic and political awareness of solar capacities, innovation, and developments.
- Bottom-up Flourishing: Sense of camaraderie, stewardship, and a strengthening of cultural character within local communities and neighborhoods.
- Cross-sectoral Collaboration: An ethical, transparent, and iterative communication and design process across all involved.
- Context Embeddedness: Sensitivity towards local and sustainable understanding of space and place in urban design and planning.
- Aspirations: Hope, motivation, determination, plurality, flexibility in the creative process (in solar design generally, and specifically within expanding the craft of a solar designer).

Indeed, the renewables industry is growing and evolving at such a fast pace that beyond making sense of solar in context, it is important to interrogate its presumed ideals alongside local, contextual sustainability goals and efforts. The workshop with youths shows – while situated to the context of [anonymised] – that misalignment towards the sustainability transition is not solely a phenomenon plaguing working adults, and that perhaps even more importantly, attention should be shifted to younger generations in honing their increased agency and power towards the renewables transition. This buffers, to some degree, the potential uncertainty in the becoming of renewables technology of the future – that they would be created in a just, resilient, and adaptable way. The workshop indicated that youths' frustration towards the current renewables transition might require the emerging role of a solar designer to not only consider plural, multi-scalar, and holistic approaches, but also necessarily embedded in an inter-generational perspective – a role that reconciles situated knowledges of younger and older social groups.

To conclude, the role of the solar designer is not stagnant; it requires proactive, empathetic, introspective and phenomenological grounding on what solar is and what it could be, to both human and non-human lives. But more importantly, bravery to genuinely reconcile with how – and if it fits at all – into the interpreted sustainability transition, values, and ideals, of a situated context. This article does not attempt to provide an encompassing overview of what it means to be a solar designer, but offers insights into the present and future possibilities of its role in increasingly uncertain times. With hope, I continue to question the techno-rational design paradigm in solar design (and renewables as a whole), towards one that is more critical and reflective, building on pre-existing research that tends to this matter. These contentions might extend to wider discussion and debates on other renewables, bringing to fore a central query on the temporal presence and form of renewables in human living environments – that ideally would maintain its humanness over time.

Consent for publication

NA.

Ethics approval

NA.

AI statement

No form of AI in any way was used in the planning, writing or revision of this document.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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